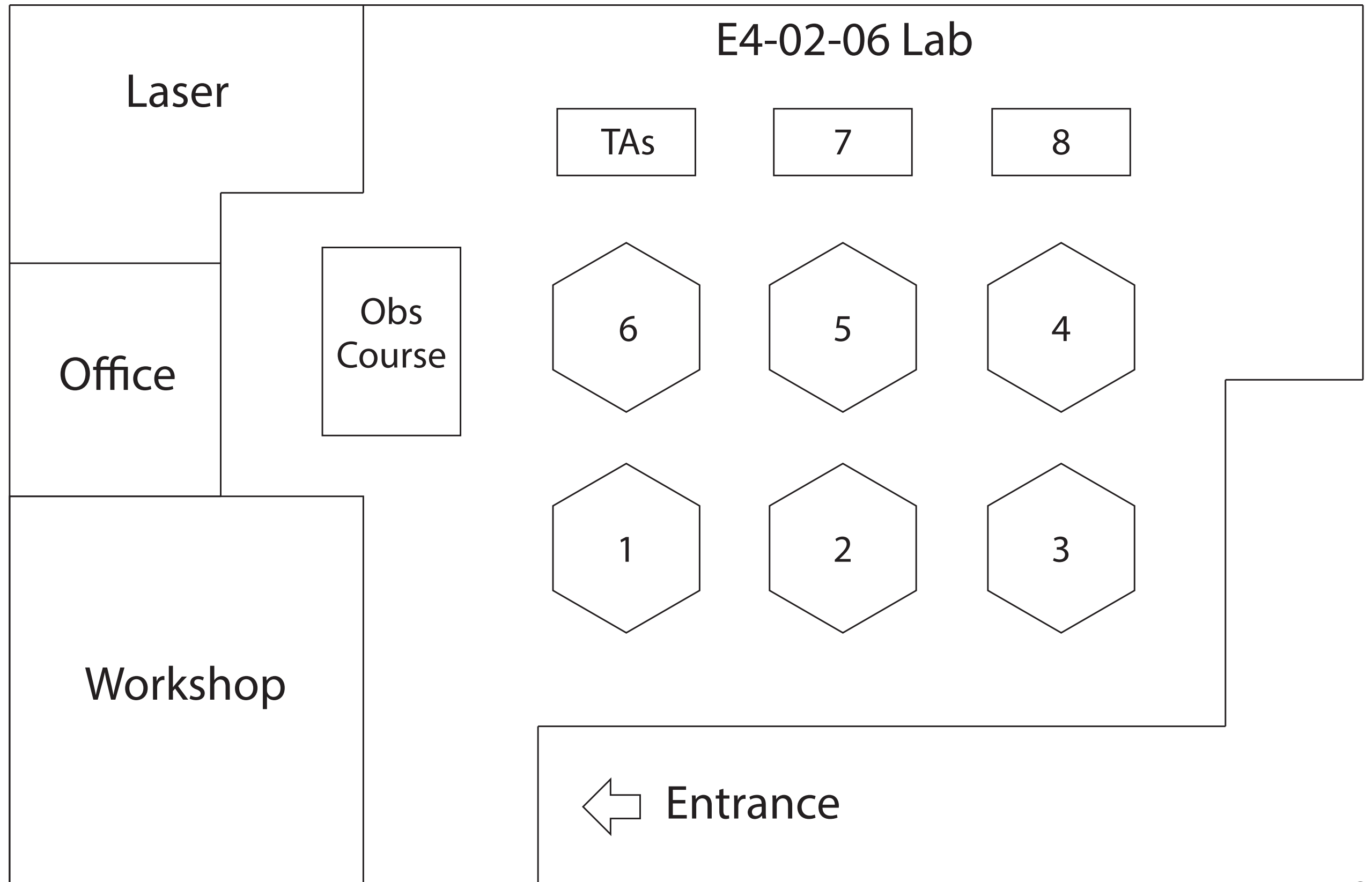


EG1311: Design & Make

Lab Safety

Dr. Jason S. Ku

E4-02-06 Lab



Lab Rules



- No **food** or **drink** in lab
- No open **shoes** (closed-toe and back)
- No shorts/skirts (**pants**, long skirt ok)
- Do not sit on the tables



Lab Safety Video

Laboratory Safety & Health Orientation

EG1311 Lab Risks

- Cut from use of handheld pen knife
- Burn from use of hot glue gun
- Shock from use of low-voltage electronics

Risk Level		Likelihood		
		1	2	3
Severity	1	1	2	3
	2	2	4	6
	3	3	6	9

**Please fill out
Safety Form**

1-2 Low Risk, Additional Controls Not Required

3-5 Consider Additional Risk Controls

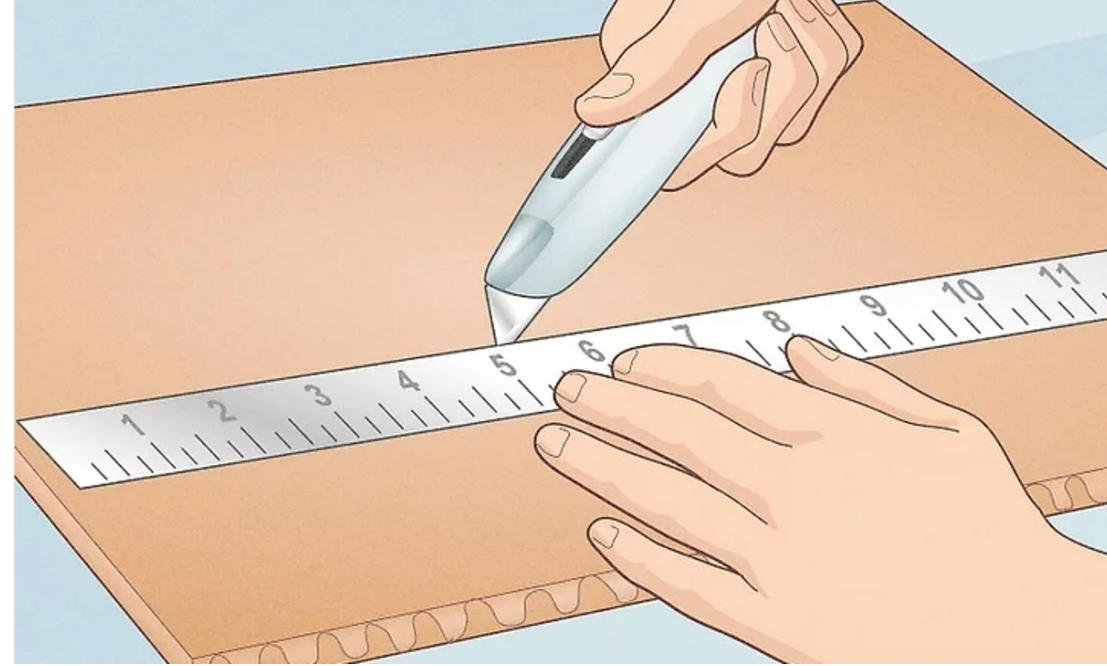
6-9 Additional Risk Controls Required

Pen Knife Safety



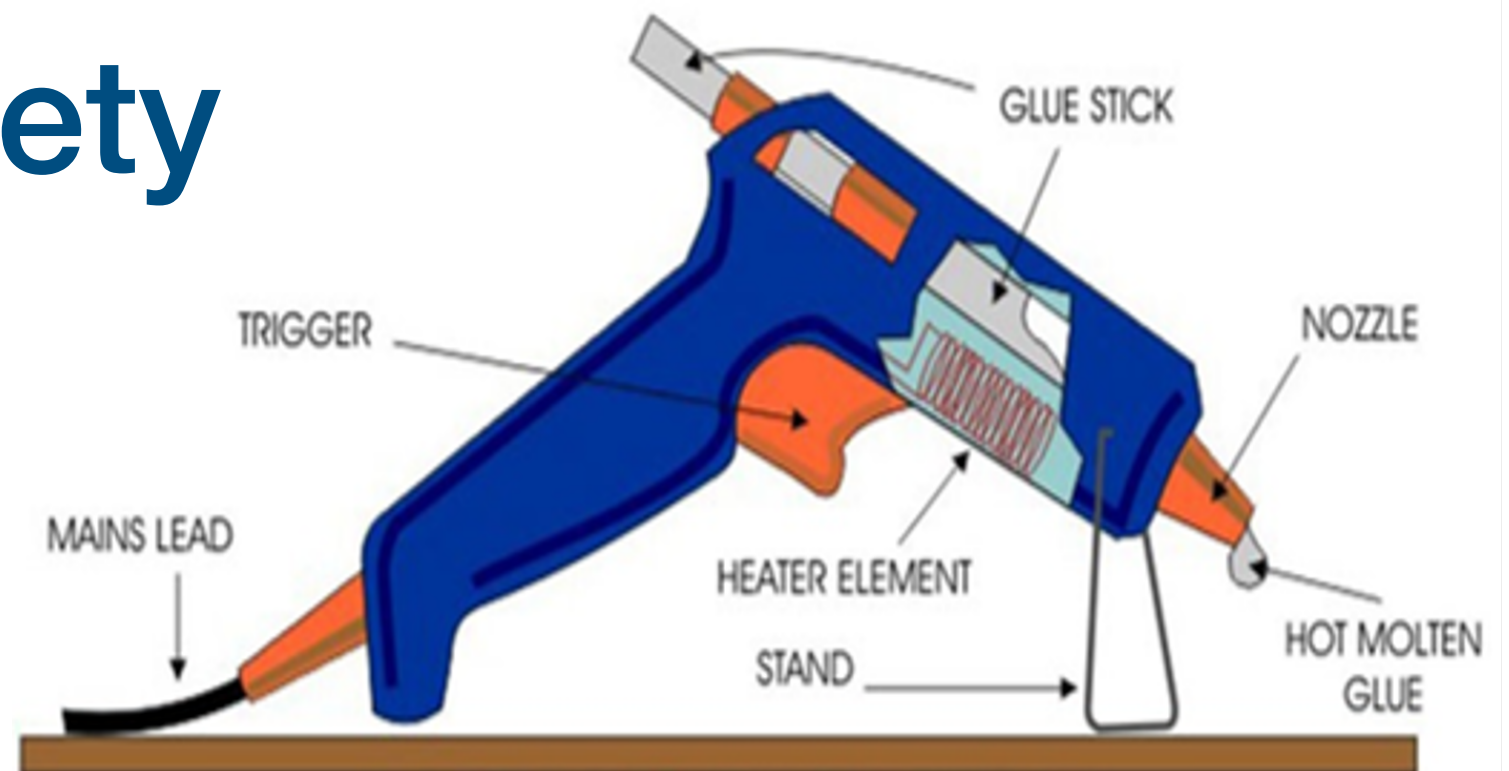
- Always cut on a **cutting surface** or a **cutting mat**
- Ensure cutting mat is **stable** and **unobstructed**
- Ensure pen knife is in **good condition** before use
(not damaged, nothing loose)
- **Lock the blade** in place before cutting
- **Hold handle securely** to maintain control
- Hold knife at **45-degrees** and apply firm pressure

Pen Knife Safety



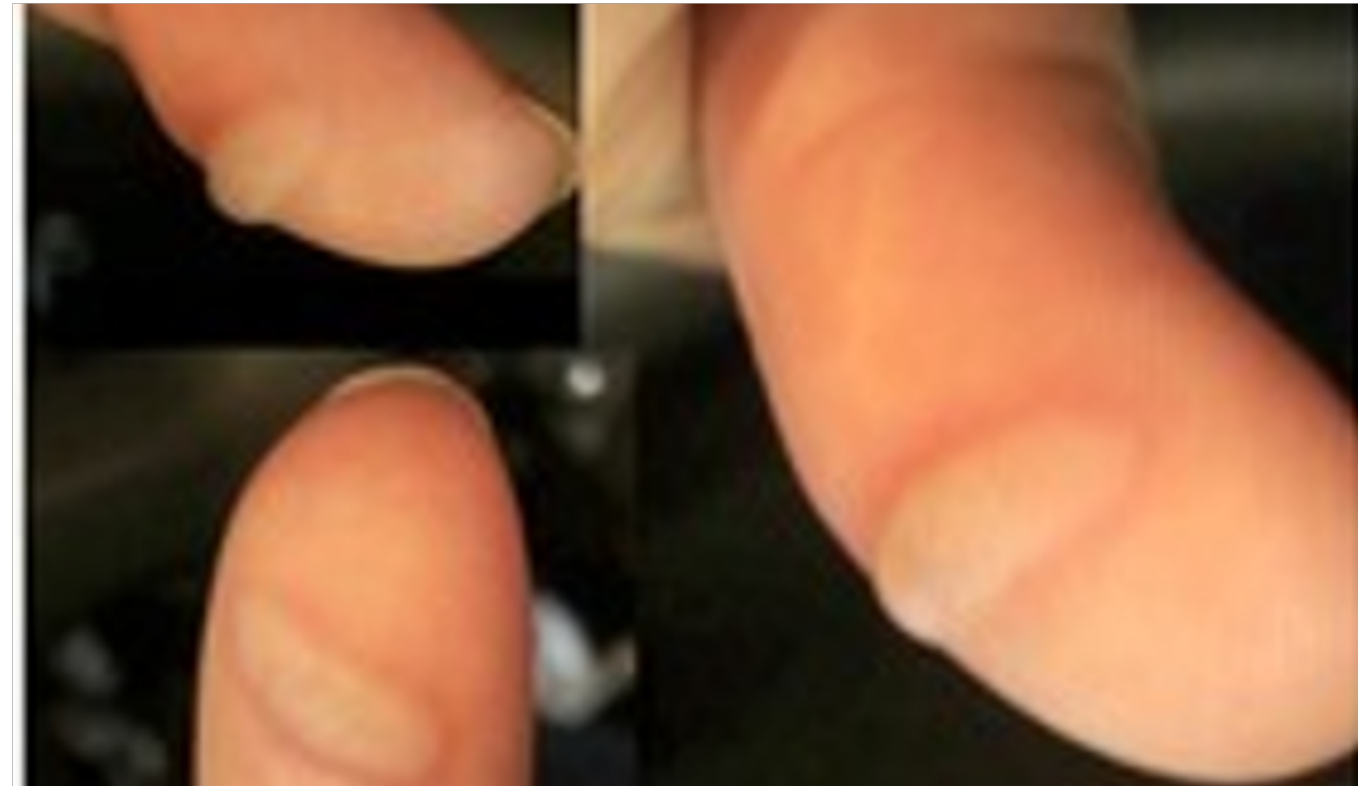
- **Use a ruler** to make clean, straight cuts
- Don't force the blade, be **controlled & moderate**
- **Take your time** and cut slowly
- Cut **away from your body** to avoid injury
- Ensure **fingers** are not in the path of the blade
- **Fully retract** the blade when not in use
- **Do not engage in horseplay** with your knife

Glue Gun Safety



- Glue gun needs **5 minutes to heat up** before use
- Glue gun should **stand upright** when not in use
(Not laid on its side)
- Always keep glue gun pointed in a **safe direction**
- Apply **light pressure** to slowly release hot glue
- Do not use **excessive force** on the trigger

Glue Gun Safety



- **Do not touch** hot glue gun nozzle or melted glue
- **Avoid contact** with the skin
- Do not leave **unattended** while plugged in
- When finished, **turn off** glue gun and **let cool**

Electrical Safety



- Do not plug **2-pin plug** directly into lab outlet
(always use **3-way Multi adaptor**)
- Ensure batteries are **not damaged** before use
- **Do not tape or glue** anything to the batteries
- **Unplug** batteries from holders when not in use
- Do not **short** + and - ends of battery
- If component becomes hot, **unplug power ASAP**

Safety Form

Quiz instructions

Please fully read and fill out the following safety form.



Question 1

0 pts

Contact Information

- Personal phone:
- Emergency contact name:
- Emergency contact phone:



Question 2

2 pts

EG1311 Project Risks

- Mechanical:
 - Potential mild cut from use of handheld pen knife.
 - Potential mild burn from use of hot glue gun.
- Electrical:
 - Potential mild shock from use of rechargeable batteries.
- Controls:
 - Students are instructed in use of, and possible dangers associated with lab supplies as well as precautions which must be observed.

☐ I have watched the video: "Introduction to Laboratory Safety & Health in NUS"

☐ I have been briefed on the risk assessments and controls of EG1311.



Question 3

1 pts

Guidelines for Appropriate Use of Laboratory Facilities

1. Be responsible for your own safety and proper handling of your project setup. Think of the impact on others as a result of your action.
2. Wear proper attire in the laboratory. **No sandals, slippers or any open-toed shoes** are allowed. **No shorts or skirts** are allowed (legs must be covered). Observe additional safety measures required at other areas.
3. **Do not work alone in the laboratory after office hour.**
4. **Strictly NO FOOD is allowed in the laboratory.**
5. No Visitors are allowed in the laboratory without permission from the laboratory management/Head of Department.
6. Dispose wastes (e.g. chemicals and biologicals) in accordance to safety guidelines where appropriate.
7. Sign your attendance on the **LOG BOOK** when accessing the laboratory after office hours.
8. No operation of machines/equipment is allowed before any briefing on respective safe machine operation procedure. Follow the Standard Operating Procedures (SOPs).
9. **Report any incident or accident immediately to the laboratory staff during office hours and to the campus security officer/faculty safety officer after office hours. Check safety notice board for the contact numbers.**

☐ I agree to abide by the above Guidelines for Appropriate Use of Laboratory Facilities.



Question 4

2 pts

Declaration

- ☐ I shall not hold the University responsible or liable in the event of death, injury, disability or any mishaps while participating in the laboratory activities.
- ☐ I have read, understood and hereby acknowledged receipt of an exact copy of this Registration form.

Saved at 14:11

Submit quiz

EG1311: Design & Make

Lab Safety

Dr. Jason S. Ku